

Synoptic CB: Porewater DOC, TN

September 2022 Samples

2026-06-17

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##Setup - Change things here & write any notes

```
#identify section
cat("Setup Information")
```

Setup Information

```
##### Run information - PLEASE CHANGE
Date_Run = "11/3/2022" #Date that instrument was run
Run_by = "Stephanie J. Wilson" #Instrument user
Script_run_by = "Zoe Read" #Code user
run_notes = "Standard Curve assessed manually on the instrument. Some sample IDs are missing from metadata"
samples <- c("GCrew", "GWI", "MSM", "SWH") #whatever identifies your samples within the same names

samples_pattern <- paste(samples, collapse = "|")
#samples_pattern <- "GCW" #use this instead of the line above if you have only one site code
chks_name = "Chk" #what did you name your check standards?

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name = c("Raw Data/TOCTN_COMPASS_Synoptic_TCTN_202209_1.txt" , "Raw Data/TOCTN_COMPASS_Synoptic_TCTN_202209_1.txt")
```

```

#file path and name for raw all peaks file
# raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202505_allpeaks.txt"

#file path and name of processed data file
processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202209.csv"

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2022.csv"

#qac log file path for this year
#Log_path = "Raw Data/COMPASS_Synoptic_TOCTN_QAClog_2022.csv"

```

```
##Set Up Code
```

```
##Read in metadata and create similar sample IDs for matching to samples
```

0.1 Import Data Functions

0.2 Import Sample Data

```
## Import Sample Data
```

```
## New names:
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## # A tibble: 6 x 4
```

	sample_name	npoc_raw	tdn_raw	run_datetime
	<chr>	<dbl>	<dbl>	<chr>
## 1	22_MSM_202209_WC_LysC_10cm	21.0	2.25	11/3/2022 7:21:47 PM
## 2	23_MSM_202209_WC_LysC_20cm	38.2	3.59	11/3/2022 7:49:29 PM
## 3	25_MSM_202209_SW_A	7.67	0.589	11/3/2022 9:41:35 PM
## 4	26_MSM_202209_SW_B	7.96	0.657	11/3/2022 10:08:45 PM
## 5	27_MSM_202209_SW_C	7.79	0.649	11/3/2022 10:32:59 PM
## 6	28_GCrew_202209_UP_LysA_20cm	18.6	0.658	11/3/2022 10:57:34 PM

```
##Fix sample naming
```

```

#remove number from sample ID
dat_raw <- dat_raw %>%
  mutate(sample_name = str_remove(sample_name, "[^_]*"))

#fix entry error
dat_raw <- dat_raw %>%
  mutate(sample_name = gsub("Lsy", "Lys", sample_name))

#want date to be at the beginning for now
#MSM_202209_WC_LysC_10cm --> 202209_GWI_UP_LysA_10cm
dat_raw <- dat_raw %>%

```

```
separate(sample_name, into = c("Site", "Date", "Zone", "Replicate", "Depth")) %>%
mutate(sample_name = paste(Date, Site, Zone, Replicate, Depth, sep = "_")) %>%
mutate(sample_name = gsub("_NA", "", sample_name)) %>%
select(-c("Site", "Date", "Zone", "Replicate", "Depth"))
```

```
## Warning: Expected 5 pieces. Missing pieces filled with 'NA' in 17 rows [3, 4, 5, 23, 24,
## 25, 49, 50, 70, 73, 74, 75, 94, 95, 96, 120, 121].
```

```
##fix GCW designation
dat_raw <- dat_raw %>%
  mutate(sample_name = gsub("GCrew", "GCW", sample_name))
```

0.3 Assessing Standard Curves - assessed manually on the instrument

0.4 Assess Check Standards

```
## Assess the Check Standards
```

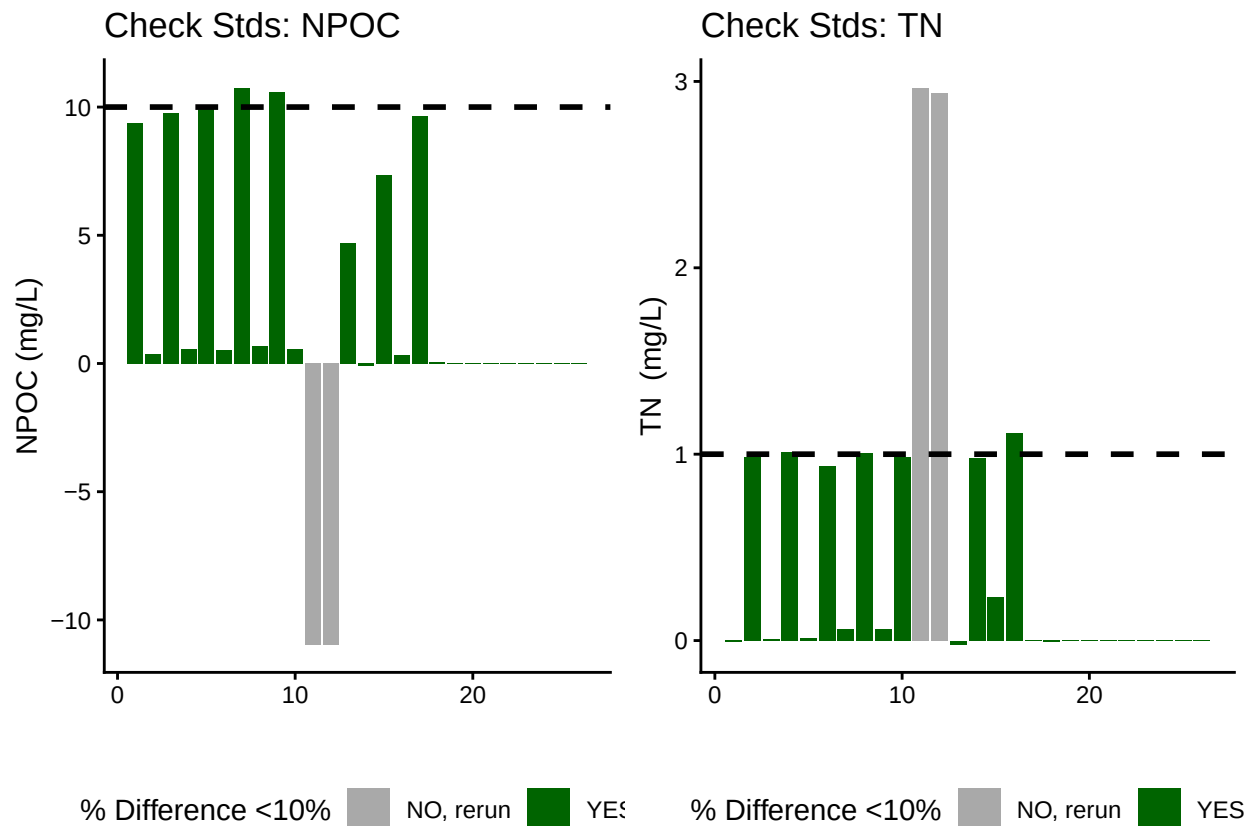
```
## New names:
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## [1] "Carbon CHECK STANDARD RSD TOO HIGH - REASSESS"
```

```
## [1] "Nitrogen CHECK STANDARD RSD TOO HIGH - REASSESS"
```



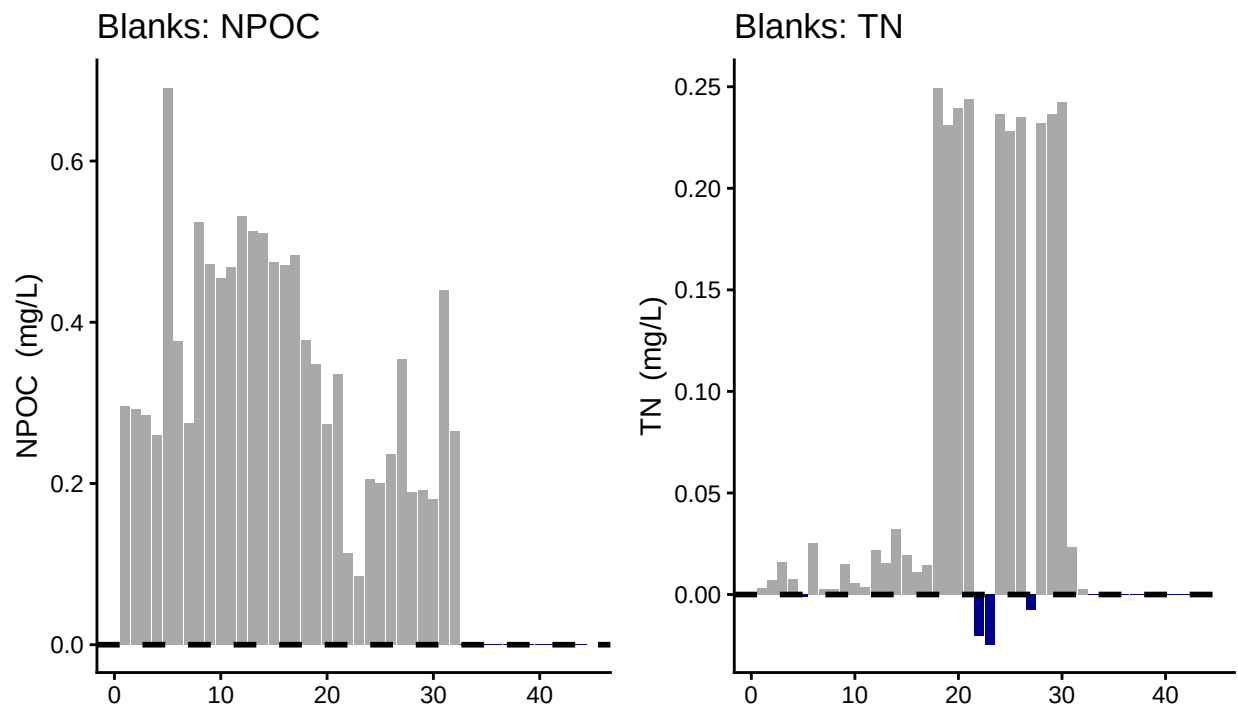
```
## [1] ">50% of Carbon Check Standards are within range of expected concentration"
## [1] ">50% of Nitrogen Check Standards are within range of expected concentration"
```

0.5 Assess Blanks

```
## Assess Blanks
```

```
## New names:
## New names:
## * ' ' -> '...14'
```

```
## [1] "<50% of Carbon blanks are lower 25% quartile of samples - REASSESS"
## [1] "<50% of Nitrogen blanks are lower 25% quartile of samples - REASSESS"
```



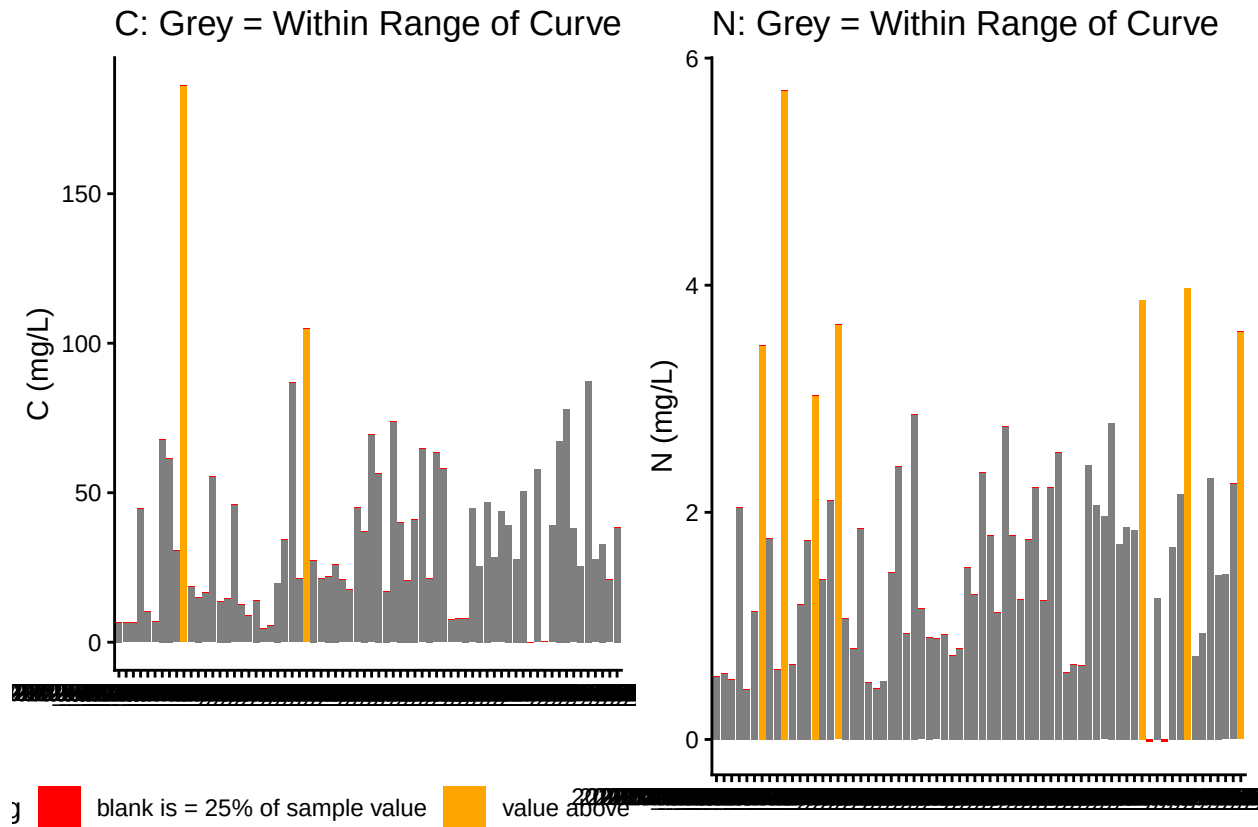
Blank Conc <25% Quartile Samples NO, rel Blank Conc <25% Quartile Samples NO, rerun

```
## carbon blanks:
## [1] 0.2539468
## nitrogen blanks:
## [1] 0.05795932
```

0.6 Assess Duplicates - no duplicates run

0.7 Sample Flagging - Are samples Within the range of the curve?

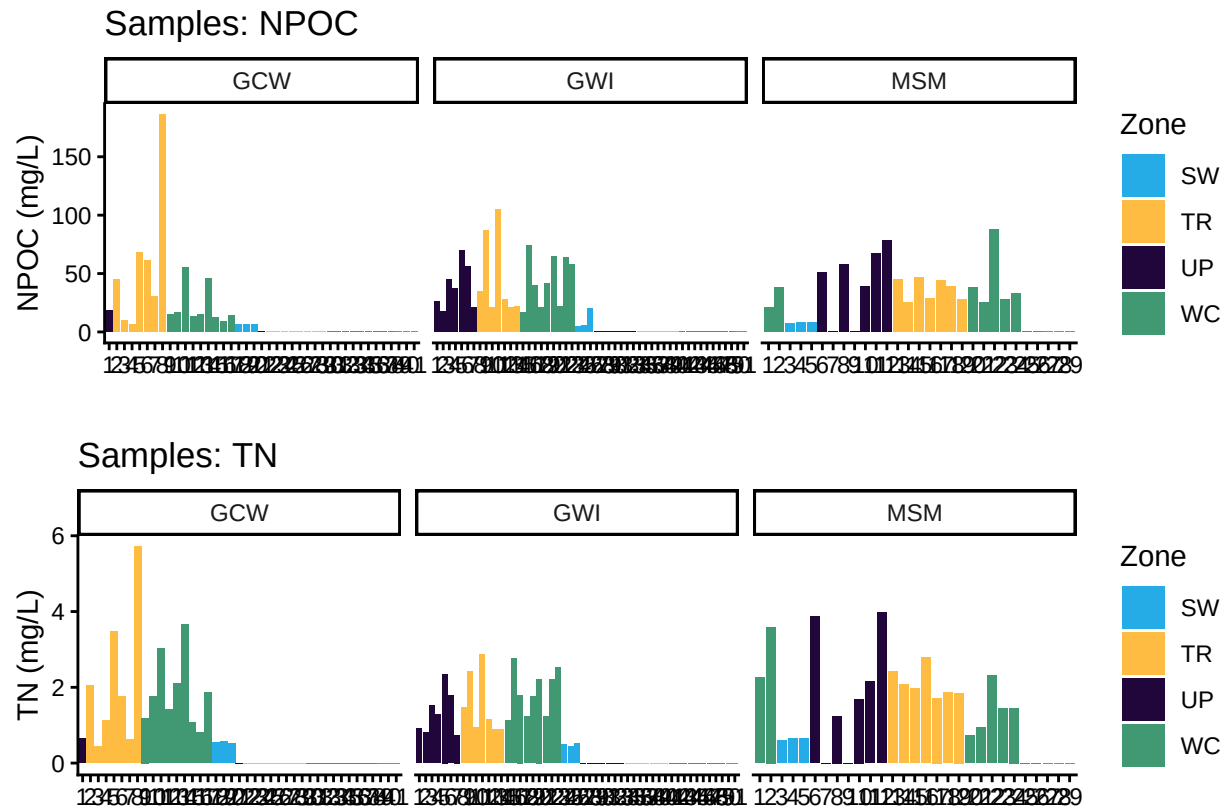
Sample Flagging



0.8 Visualize Data by Plot

Visualize Data

```
## Warning in rbind(c("202209", "MSM", "WC", "LysC", "10cm"), c("202209", "MSM", :  
## number of columns of result is not a multiple of vector length (arg 3)
```



0.9 Convert data from mg/L to uMoles/L

0.10 Check to see if samples run match metadata & merge info

```
## Check Sample IDs with Metadata

## Some sample IDs are missing from metadata.

## [1] "202209_GCW_UP_LysA_20cm" "202209_GCW_TR_LysB_45cm"
## [3] "202209_GCW_TR_LysC_10cm" "202209_GCW_TR_LysC_20cm"
## [5] "202209_GWI_UP_LysB_45cm" "202209_GWI_UP_LysC_20cm"
## [7] "202209_GWI_UP_LysA_45cm" "202209_MSM_TR_LysC_45cm"
```

0.11 Export Processed Data

```
## Export Processed Data

## # A tibble: 6 x 21
##   Project      Region Site Zone Replicate Depth_cm Sample_ID Year Month Day
##   <chr>        <chr> <chr> <chr> <chr>      <chr>      <int> <int> <int>
## 1 COMPASS: Sy~ CB    MSM  WC    LysC      10      202209_M~ 2022    9    18
## 2 COMPASS: Sy~ CB    MSM  WC    LysC      20      202209_M~ 2022    9    18
## 3 COMPASS: Sy~ CB    MSM  SW     A          202209  202209_M~ 2022    9    18
```

```
## 4 COMPASS: Sy~ CB      MSM  SW   B      202209  202209_M~ 2022    9    18
## 5 COMPASS: Sy~ CB      MSM  SW   C      202209  202209_M~ 2022    9    18
## 6 COMPASS: Sy~ CB      GCW  UP   LysA    20      202209_G~  NA    NA    NA
## # i 11 more variables: Time <lgl>, Time_Zone <lgl>, npoc_mgL <dbl>,
## #   npoc_uM <dbl>, npoc_flag <chr>, tdn_mgL <dbl>, tdn_uM <dbl>,
## #   tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>, Field_notes <chr>
```

```
#end
```